MIL-R-28966(MC)
29 January 1971
SUPERSEDING
See Section 6.

MILITARY SPECIFICATION

REPRODUCTION SET, SPIRIT PROCESS

1. SCOPE

1.1 This specification covers the requirements for one type reproduction set consisting of a hand-operated, spirit-process duplicator, a military field chest, and an expendable supply set.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-D-825	- Duplicating Fluid, Direct Process, (Spirit Type)
P-H-0031	- Handcleaner, Duplication Ink, Hand Cleaner Grease
QQ-S-698	- Steel, Sheet And Strip, Low-carbon
PPP-B-636	- Box, Fiberboard
PPP-B-640	- Box, Fiberboard, Corrugated, Triple-wall

MILITARY

MIL-P-116	- Preservation, Methods Of
MIL-T-152	- Treatment, Moisture- And Fungus-resistant, Of Communications, Electronic, And Associated Electrical Equipment
MIL-V-173	- Varnish, Moisture- And Fungus-resistant, (For Treatment Of Communications, Electronic, And Associated Equipment)

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MIL-C-12044 - Chest, Plywood

STANDARDS

MILITARY

MIL-STD-129 - Marking For Shipment And Storage
MIL-STD-130 - Identification Marking Of U.S. Military Property

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards For Federal Service
PS 1-66 - Softwood Plywood

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

AMERICAN IRON AND STEEL INSTITUTE

Steel Products Manual - Carbon Sheet Steel, Coils and Cut Lengths

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East Forty-Second Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 First article. The first article is defined as a production model identical to the product that the contractor will subsequently deliver in fulfillment of the contract. As soon as practicable after award of the contract or on a date specified in the contract, and prior to submission of any units for final acceptance, the contractor shall furnish a first article for examination and test to determine conformance to this specification. Examination and testing shall be as specified herein. Approval of the first article by the procuring activity shall not relieve the contractor of his obligation to supply a product which conforms to this specification. Any changes or deviations of production units from the first article shall be subject to the approval of the contracting officer. The first article, when approved, may be included in the quantity to be shipped in the final shipment.

3.2 Materials. All materials used in the fabrication of the reproduction set shall be new and of sufficient quality and strength to meet normal use

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shall be avoided whenever possible. All surfaces shall be smooth and free from burrs or sharp edges. Molding fins and excess metal shall be removed from castings.

3.3 Design and construction. The reproduction set shall consist of a hand-operated spirit-process duplicator, a military field chest and an expendable supply set. All parts shall be new, of first class quality, and of the latest approved design. All parts shall be of such size, material and strength as to sustain properly the maximum allowable work-loads and capacities imposed upon them, with an adequate factor of safety, maximum working efficiency and minimum wear.

3.3.1 Duplicator. The duplicator shall be designed and constructed in accordance with standard commercial practice for this type of equipment. The duplicator shall be mounted on the machine tray and be capable of operating while sitting on top of the military field chest. The duplicator shall be capable of reproducing a minimum of 200 copies per master. There shall be no deterioration of quality in the first 50 copies. Deterioration after 50 copies and up to 200 copies shall be reasonable but shall not include loss of clarity or sharpness of detail. The duplicator shall have a 14-inch by 17-inch mastersheet capacity and a 13-1/2-inch by 16-1/2-inch duplicating area. In addition, the duplicator shall be hand-operable and shall include a gravity-flow fluid system, a copy positioning device, and a provision for adjusting roller impression.

3.3.1.1 Fluid system. The fluid system shall be gravity fed, with a capillary fluid feed through a wick moistener and wipers, and shall operate without priming. The fluid tank shall have a minimum capacity of 1 quart and shall contain a positive fluid shutoff.

3.3.1.2 Copy positioning device. The copy positioning device shall be graduated for 3 inches up or down, with hairline registration, and shall be actuated by rotating a control knob.

3.3.1.3 Roller pressure adjustment. Roller pressure adjustment shall be accomplished by rotating a control knob.

3.3.2 Military field chest. The military field chest (see Figure 1) shall be fabricated of exterior type plywood, 1/2-inch thickness, grade AC or better, conforming to PS 1-66. All joints shall be dadoed approximately 1/8 inch, nailed or stapled, and glued with waterproof glue. All exterior edges of the chest and door shall be protected by steel angles with the sizes and method of attachment in accordance with MIL-C-12044. The steel binding shall be 0.0299 inch (Manufacturer's Standard Gage No. 22 in accordance with the Steel Products Manual) and shall conform to QQ-S-698. The back shall be securely glued and nailed or stapled into the shelves at 4-inch centers. The binding angles for the top and bottom shall be continuous at the corners. The vertical angles shall abut the end angles and the last rivet shall be approximately 3/8 inch from the end.

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COMPARTMENT	CONTENTS	QUANTITY
(A)	SPIRIT MASTER PAPER (100-SHEET PKG)	1
	CARBON DUPLICATING MACHINE PAPER (25-SHEET PACKAGE OF EACH OF FIVE COLORS: PURPLE, RED, GREEN, BLUE, AND BLACK)	1 PKG EACH COLOR
(B)	POURING SPOUT	1
	DRAIN BOTTLE	1
	CORRECTION BLADE	1
	FEED ROLLERS	1 SET
	SIDE GRIPPERS	1
	HAND CLEANER (8-OUNCE TUBE)	1
(C)	DUPLICATING FLUID (1-GALLON CAN)	2
	SPARE GALLON CAN	1
(D)	OVERLAY PAPER	2 REAMS
(E)	SPIRIT PROCESS DUPLICATOR	1

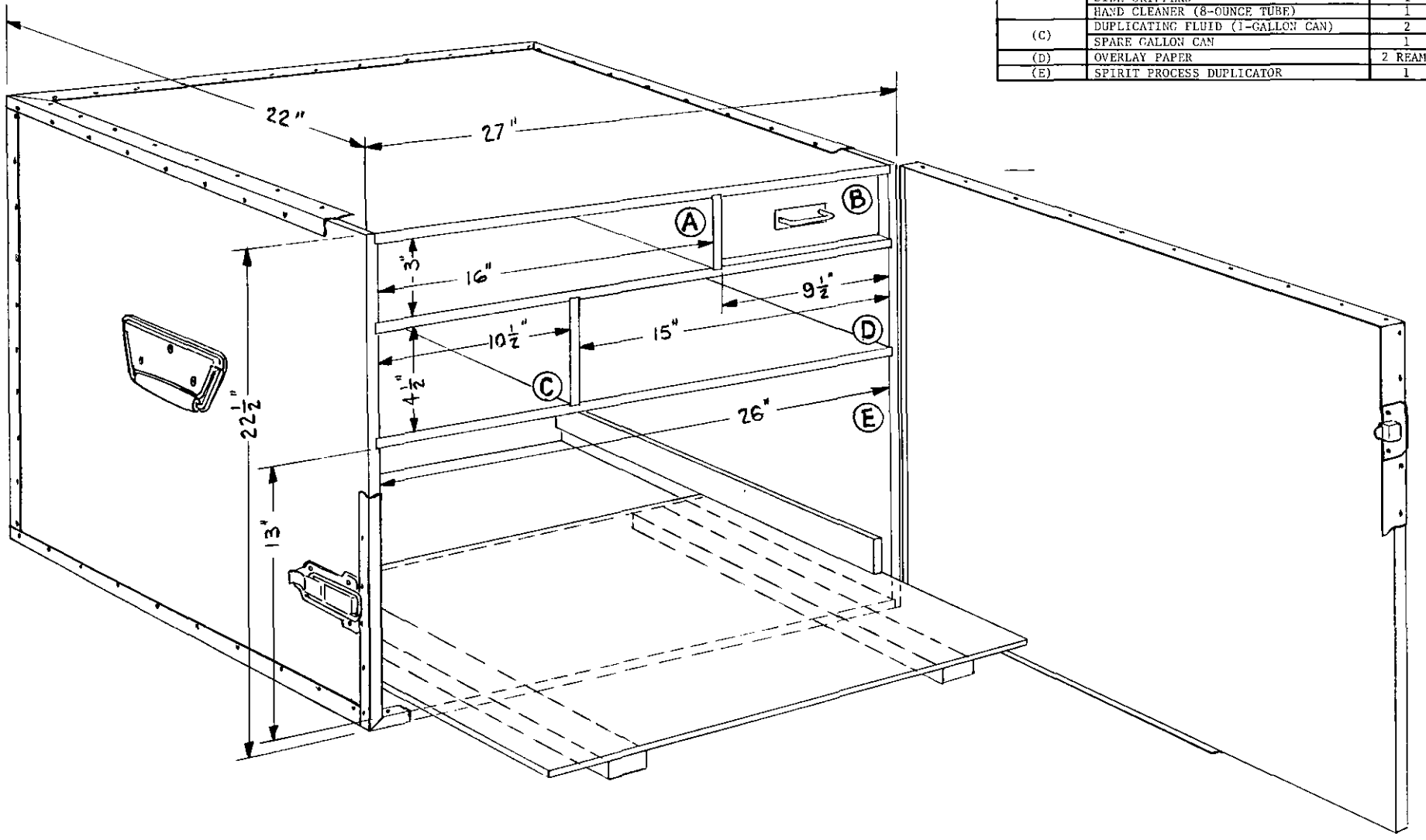


Figure 1. Military Field Chest, Dimensions and Contents of

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3.3.2.1 Door. The door shall be protected and the edge of the chest shall be bevelled to accommodate the protruding metal lip.

3.3.2.2 Hinges. The door shall be mounted with slip-pin hinges to allow ready removal. The hinges shall not be less than 2 inches, each half being rivetted with three rivets in a manner which assures a tight closure.

3.3.2.3 Draw bolts. Two draw bolts shall be attached and mounted, opposite the hinges, in a manner which assures a tight closure.

3.3.2.4 Handles. The military field chest shall be provided with a handle on each side, mounted 14 inches from the bottom and attached with sleeve nuts.

3.3.2.5 Accessory drawer. The accessory drawer shall be constructed of 1/4-inch plywood of appropriate size to fit into the compartment. The drawer handle shall be a simple pull-type handle attached so as to secure the drawer snugly when the door is shut.

3.3.2.6 Machine tray. The machine tray shall be 1/4-inch plywood of appropriate size to readily slip under two retaining strips when the duplicator is mounted thereon. The cut edges shall be smooth so as not to cause injury to personnel when the duplicator is on top of the military field chest.

3.3.2.7 Skids. The skids shall be of a suitable hardwood, 1 inch by 2 inches. The skids shall be of a length which will abut, but not overlap, the binding angles of the door and the back of the chest when the duplicator is set on top of the field chest. The skids shall be mounted on the machine tray so that they slightly overlap the binding angle of the chest to assure unrestricted removal from the chest when the duplicator is mounted on the tray.

3.3.2.8 Hardware. All nails, staples, rivets, and other permanent-type hardware shall be in accordance with best commercial practices and shall be of sufficient size and strength for the intended purposes. All fastening devices such as screws, pins, bolts, and similar parts shall be installed, wherever practical, with a positive means to prevent loss of tightness. All such parts shall not be swaged, peened, staked or otherwise permanently fixed. All threaded parts shall conform to H28.

3.3.3 Expendable supply set. The expendable supply set shall consist of the following:

Item	Quantity
Paper, overlay, 14 inches by 17 inches, 16 lbs., ream	2
Paper, spirit master, 14 inches by 17 inches, 60 lbs., sheet	100

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Item (cont.)	Quantity (cont.)
Paper, carbon duplicating machine, designed for direct process use, 14 inches by 17 inches, sheets	
Purple	25
Red	25
Green	25
Blue	25
Black	25
Can, screw cap, 1 gallon, each	1
Spout, pouring, each	1
Bottle, drain, each	1
Blade, correction, each	1
Rollers, feed, set	1
Grippers, side, each	1
Duplicating fluid, direct process; high menthol; 1-gallon can; in accordance with O-D-825, type 2	2
Hand cleaner liquid; designed to remove duplicating ink; 8-ounce jar; in accordance with P-H-0031, type 1	1

3.3.4 Corrosion resistance. All components of the reproduction set shall be corrosion resistant, or shall be treated to resist corrosion.

3.3.5 Fungus control. Unless otherwise specified (see 6.2), the reproduction set shall be treated for fungus resistance in accordance with type 1 requirements of MIL-T-152 using material conforming to requirements of MIL-V-173.

3.3.6 Safety devices. The reproduction set shall be equipped with the manufacturer's normal safety devices for this type of equipment and any other safety devices or protective covering for all parts or components which present safety hazards.

3.3.7 Interchangeability. All parts or components shall be constructed to standards which will permit their replacement or adjustment without modification to any items in the reproduction set.

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3.3.8 Accessibility. All parts or components subject to wear, breakage, or distortion shall be accessible for adjustment and repair.

3.4 Performance. The reproduction set shall be capable of performance as specified herein under environmental conditions as follows.

3.4.1 Vibration. The reproduction set shall be capable of undergoing the vibration normally encountered in transport by air, land or sea.

3.4.2 Shock. The reproduction set shall be capable of undergoing the shock normally encountered in vehicular transportation through military field conditions and from being accidently dropped.

3.4.3 Temperature. The reproduction set shall be capable of withstanding the following temperatures:

(a) Storage. The reproduction set, excluding the expendable supply set, shall withstand storage in a temperature of -65°F to +155°F without damage. The expendable supply set shall withstand storage in temperatures of +33°F to +155°F.

(b) Operating. The reproduction set shall operate, without decrease in quality of copies, in ambient air temperatures from +33°F to +105°F.

3.4.4 Humidity. The reproduction set shall withstand exposure to and operate under cover in 15 to 100 percent relative humidity at air temperatures from +85°F to +105°F.

3.4.5 Rain. With the door closed and fastened, the military field chest shall not show evidence of leakage when subjected to heavy rain.

3.5 Dimensions. The military field chest dimensions shall conform to Figure 1. The duplicator shall be of such dimensions as to fit easily into the military field chest.

3.6 Weight. The duplicator shall weigh from 50 to 70 pounds.

3.7 Finish. Unless otherwise specified (see 6.2), the duplicator shall be finished and painted in accordance with the manufacturer's standard commercial practice. The exterior and interior of the military field chest shall be treated and painted in accordance with MIL-C-12044. The color shall match Marine Corps green No. 23 (see 6.3).

3.8 Identification marking. The duplicator shall be marked for identification in accordance with MIL-STD-130, using a metal identification plate permanently attached to the duplicator. The military field chest shall be marked for identification using a metal identification plate permanently attached to the side panel upon which the door hinges are mounted.

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The information to be included on the plate shall be as follows:

REPRODUCTION SET, SPIRIT PROCESS

Federal Stock No._____
Manufacturer's Part No._____
Name of Manufacturer (or Contractor)_____
Date of Manufacture_____

3.9 Instruction handbook. The contractor shall furnish an instruction handbook with each reproduction set supplied under this specification. The handbook shall contain necessary operating instructions, drawings, illustrations, parts lists and safety precautions to insure proper operation and maintenance of the reproduction set.

3.10 Workmanship. The reproduction set shall reflect the highest standards of manufacture consistent with the best commercial practices within industry. All functioning components shall operate smoothly without binding and without evidence of loose or misplaced components. All painted surfaces shall be uniformly coated, dry and free from sags, tear drops, foreign matter, chips, scratches, flaking and peeling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Preparation for delivery inspection (see 4.7).

4.3 First article inspection. The first article, supplied in accordance with 3.1, shall be subjected to all of the examinations and tests indicated in Table I. Failure of the first article to pass any of the examinations or tests shall be cause for rejection of all other units of that production lot. After first article tests have started, only adjustments made by externally available controls, without the use of tools, shall be permitted. Any other adjustments, repair, or replacement shall be considered a failure.

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4.4 Quality conformance inspection. The quality conformance inspection shall consist of the following:

(a) Individual. Each reproduction set manufactured for delivery in fulfillment of the contract or order shall be tested as specified in Table I. Any defect noted shall be cause for rejection of that unit. The unit rejected may be resubmitted for inspection after the defect that caused its rejection has been corrected. Acceptance or rejection of one unit shall not be cause for acceptance or rejection of other units.

(b) Sample. One reproduction set selected at random, under the direction of a Government representative, from each production lot of 25 or less serially-numbered reproduction sets produced on a contract or order, shall be subjected to the examination and tests indicated in Table I. The reproduction set shall pass all of the examinations and tests with a maximum 25 percent defective acceptable.

Table I. Examination and tests

Type of Inspection			Examination or test	Test paragraph
First Article	Quality Conformance Individual	Quality Conformance Sample		
X	X	X	Examination	4.5
X	X	X	Operation	4.6.1
X	X	X	Copy quality	4.6.2
X		X	Vibration	4.6.3.1
X		X	Shock	4.6.3.2
X		X	Storage temperature	4.6.3.3
X		X	Low ambient operating temperature	4.6.3.4
X		X	High ambient operating temperature	4.6.3.5
X		X	Waterproofness	4.6.4

4.5 Examination. The reproduction set shall be visually examined for the following defects. Presence of one or more defects shall be cause for rejection.

(a) Expendable supply set not as specified
(b) Material not as specified
(c) Duplicator not as specified

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(d) Military field chest not as specified
(e) Painting not as specified
(f) Fungus control not as specified
(g) Safety devices not as specified
(h) Interchangeability not as specified
(i) Accessibility not as specified
(j) Identification marking not as specified
(k) Workmanship not as specified

4.6 Tests. The reproduction set shall be tested as follows:

4.6.1 Operation. The duplicator shall be set on top of the military field chest and operated in accordance with the operating procedure in the instruction handbook. The duplicator shall produce 50 clean sharp copies from a master prepared by typing and by writing and sketching with a medium ball point pen. The master shall be prepared in three colors. The typing shall be black, the writing blue and the sketch red.

4.6.2 Copy quality. The duplicator, after being operated in the manner specified in 4.6.1 shall further produce 150 copies using the same master. The last copy (200th) shall conform to requirements in 3.3.1.

4.6.3 Environmental. The environmental tests shall be as follows:

4.6.3.1 Vibration. The reproduction set, excluding the expendable supply set, shall be rigidly mounted on a vibration machine and subjected to vibrations (simple harmonic motion) with an amplitude of 0.03 inch (0.06 inch maximum excursion) through a frequency range of 10 to 35 hertz for a period of 1/2 hour in the direction of each of the three major axes. The frequency shall be continuously varied with the entire range being traversed in approximately 1 minute. At the conclusion of the test, the reproduction set shall be examined for evidence of breakage, permanent deformation, and loosening of parts, and shall be tested for serviceability employing the tests in 4.6.1 and 4.6.2.

4.6.3.2 Shock. The reproduction set shall be dropped three times onto a concrete surface from a height of 8 inches. At the conclusion of the test, the reproduction set shall be examined for breakage, permanent deformation, and loosening of parts, and shall be tested for serviceability employing the tests in 4.6.1 and 4.6.2.

4.6.3.3 Storage temperature. The reproduction set, excluding the expendable supply set, shall be subjected to an ambient air temperature of -60°F to -65°F for a period of 72 hours and a 24-hour cycling of temperature starting with 10 hours at +90°F, followed by 5 hours of steady increase to +155°F, 4 hours at +155°F, and 5 hours of steady decrease to +90°F. After this test, the reproduction set shall be examined for physical damage and shall be tested for serviceability employing the tests in 4.6.1 and 4.6.2.

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4.6.3.4 Low ambient operating temperature. The test set shall be allowed to stabilize in an ambient air temperature of +33°F, and while at that temperature, it shall be tested employing the tests in 4.6.1 and 4.6.2.

4.6.3.5 High ambient operating temperature. The reproduction set shall be subjected to 20 hours exposure with relative humidity between 93 and 97 percent at an air temperature between +80°F and +105°F, followed by 4 hours of 100 percent relative humidity with condensation at +75°F to +80°F. Air movement and radiation shall be negligible. At the conclusion of this test, the reproduction set shall be inspected for evidence of corrosion damage and shall be tested employing the tests in 4.6.1 and 4.6.2.

4.6.4 Waterproofness. With the door closed, the reproduction set shall be subjected to a spray of water for 1 hour at the rate of 2-1/2 gallons per hour, per square foot, uniformly spread over the top of the military field case. At the conclusion of this test, the reproduction set shall show no evidence of leakage.

4.7 Preparation for delivery inspection. Preservation, packaging, packing, and marking for shipment and storage shall be visually examined to determine compliance with the requirements of Section 5. Any defects shall be cause for rejection of that unit. Any unit rejected can be resubmitted for this test when the defect has been corrected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging prior to storage within the military field chest shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. All exposed critical functioning or close tolerance metallic parts of the duplicator (e.g. friction operating guides, bars, etc.) and of the expendable supply set (e.g. spare roller bearing surfaces, etc.) shall be thoroughly cleaned using cleaning process C-1 of MIL-P-116.

5.1.1.2 Drying. The duplicator shall be dried using any of the applicable drying procedures D-1 through D-5 of MIL-P-116.

5.1.1.3 Preservative application. Immediately after drying, a preservative shall be applied in accordance with MIL-P-116.

5.1.1.4 Duplicator. The duplicator shall be secured to the tray provided within the chest.

5.1.1.5 Expendable supplies. The expendable supplies shall be packaged individually in accordance with Method IC-2 of MIL-P-116 and placed within the compartments provided within the chest.

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5.1.2 Level C. The duplicator and expendable supplies shall be packaged in accordance with the supplier's normal practice provided such practice assures adequate protection against corrosion and physical damage during shipment to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The field chest with all items cushioned and in their appropriate places, as specified in 5.1.1, shall be placed in a container conforming to PPP-B-640, class 2, style E. Closure shall be in accordance with the appendix to PPP-B-640.

5.2.2 Level B. The military field chest with all items cushioned and in place as specified in 5.1.1 shall be placed in a container conforming to PPP-B-636, style RSC, type CF, class Domestic, variety DW, grade 275. Closure shall be in accordance with the appendix to PPP-B-636.

5.2.3 Level C. The reproduction set shall be packed in accordance with contractor's normal practice providing such practice assures adequate protection against climatic and physical damage during shipment to the first receiving activity.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The reproduction set is intended to be used for reproducing, by spirit process, overlays, sketches, training aids and similar items under military field conditions.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) When test kit shall be painted other than as specified in 3.7.
(c) When fungus resistance treatment shall be other than as specified in 3.3.5.
(d) Preservation, packaging, and packing requirements (see 5.1 and 5.2).

6.2.2 Contract data requirements. Data conforming to Marine Corps Data Item Descriptions DI-M-4707 and DI-T-4902 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD 1423 included in the contract.

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6.3 Color. Color chips for Marine Corps green, No. 23, may be obtained from Commanding General, Marine Corps Supply Activity (Code 843D), Philadelphia, Pennsylvania 19146.

6.4 First article instruction. The procuring activity shall specify the date and the location to which the first article is to be furnished for examination and test. Records of the first article inspection shall be kept complete by the supplier and shall be made available to the Government upon request.

6.5 Supersession data. This specification includes the requirements of Purchase Description PD 843-4-68 (MCSA), Reproduction Set, Spirit Process, dated 11 April 1968.

Preparing Activity
Navy-MC
Project No. 3610-N231

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